CIE AS PHYSICS (9702) • QUESTION PAPER

8.1 Stationary Waves

Syllabus Section: 8. Superposition **Paper Type:** Structured Questions (Paper 2) **Total:** 8 Questions (84 Marks)

Question 1 Easy [8 marks]

Part (a) [2 marks]

State the principle of superposition.

Part (b) [2 marks]

State **two** necessary conditions for a stationary wave to be set up on a string.

Part (c) [2 marks]

A stationary wave is made up of nodes and antinodes.

State what is meant by:

(i) a node

[1]

(ii) an antinode.

[1]

Part (d) [2 marks]

Fig 1.1 shows a stationary wave on a string which is fixed at both ends.

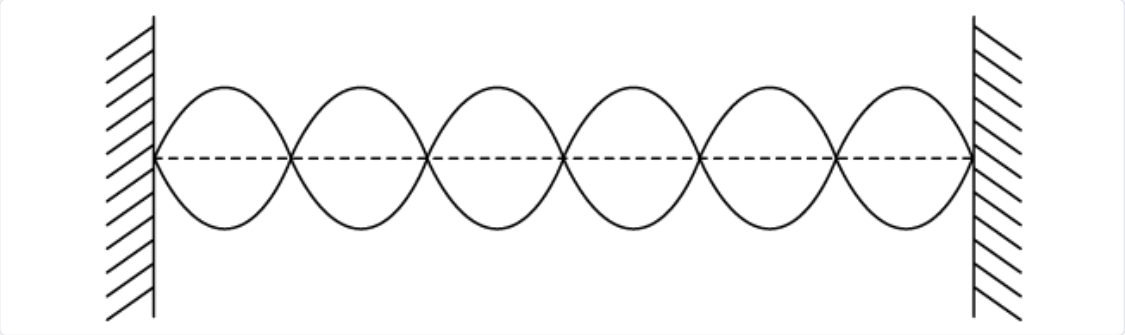


Fig. 1.1

On Fig. 1.1, label the nodes (N) and antinodes (A).

Question 2 Easy [12 marks]

Part (a) [2 marks]

Explain the difference between energy transfers in progressive waves and stationary waves.

Part (b) [3 marks]

Fig 1.1 shows a violin string. One way to produce a musical note is to pull the centre of the string to one side and then release it quickly. This produces a stationary wave.

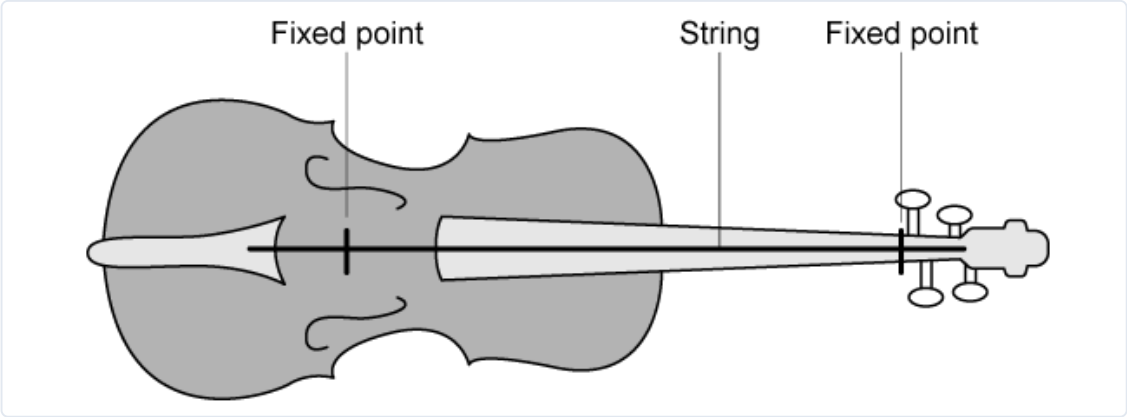


Fig. 1.1

Explain why a stationary wave is formed on the string.

Part (c) [4 marks]

Fig. 1.1 shows a clarinet producing notes of different pitch. Each note corresponds to a stationary wave of a different number of wavelengths, or half wavelengths.

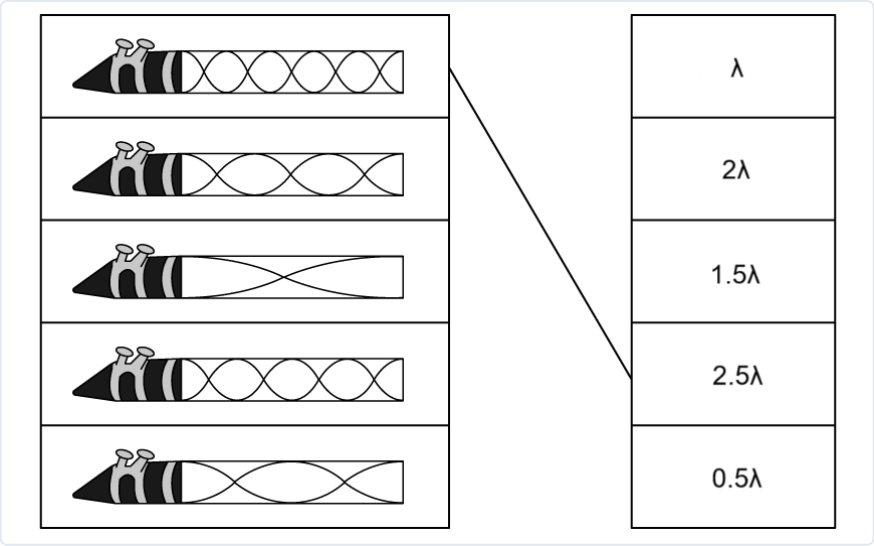


Fig. 1.1

On Fig 1.1, draw lines from each stationary wave formed in the clarinet to its corresponding wavelength.

Part (d) **[3 marks]**

State and explain the number of wavelengths that would produce the highest note.

Question 3 Easy [9 marks]

Part (a) [2 marks]

Describe the phase difference between two points on

(i) a progressive transverse wave

[1]

(ii) a stationary wave.

[1]

Part (b) [3 marks]

Fig. 1.1 shows four progressive transverse waves.

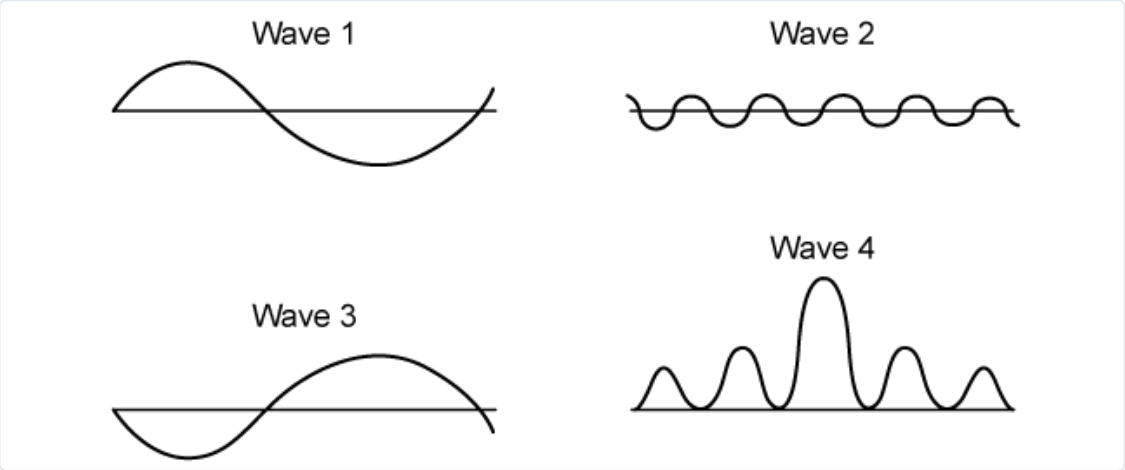


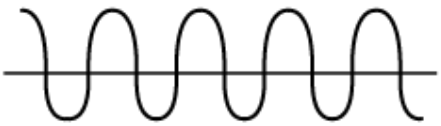
Fig. 1.1

State the pair of waves from Fig. 1.1 which superpose to create the following

(i) _____

[1]

(ii)



[1]

(iii)



[1]

Part (c) [2 marks]

Fig. 1.2 shows a stationary wave on a stretched guitar string of length 0.58 m when it is plucked.

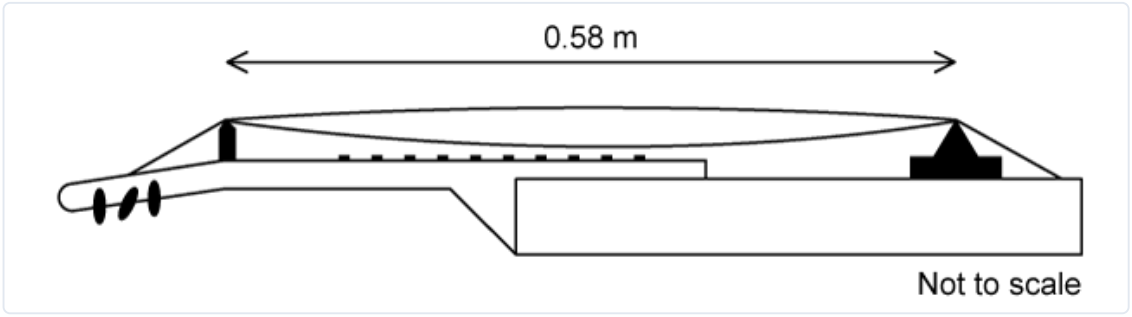


Fig. 1.2

Determine the wavelength of the stationary wave.

Part (d) [2 marks]

The frequency of the note played on the guitar string is 276 Hz.

Calculate the speed of the stationary wave in Fig. 1.2.

Question 4 Medium [3 marks]

Part (a) [3 marks]

(i) By reference to the direction of transfer of energy, state what is meant by a transverse wave.

[1]

(ii) State the principle of superposition.

[2]

Question 5 Medium [13 marks]

Part (a) [4 marks]

A stationary wave on a string is shown in Fig. 1.1.

Explain how the wave is formed, referring to the principle of superposition in your answer.

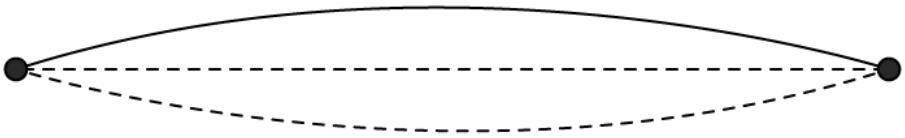


Fig. 1.1

Part (b) [1 mark]

On Fig. 1.1, draw the stationary wave that would be formed on the string in part **(a)** with two more nodes and two more antinodes.

Part (c) [5 marks]

Fig. 1.2 shows the appearance of a stationary wave on a stretched string at one instant in time.

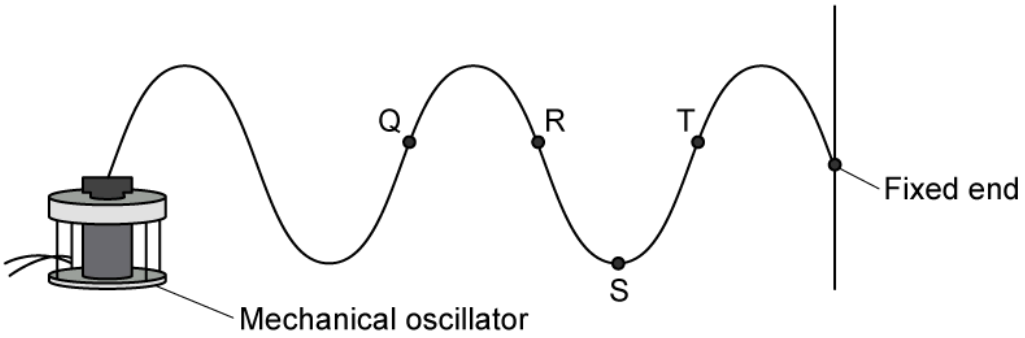


Fig. 1.2

Mark clearly on the diagram

(i) the equilibrium position,

[1]

(ii) at least two nodes by labelling them **N**, and two antinodes by labelling them **A**,

[2]

(iii) the direction in which points **Q**, **R**, **S** and **T** are about to move.

[2]

Part (d) **[3 marks]**

For the wave in Fig. 1.2 the frequency of vibration is 180 Hz and the speed of the waves along the string is 60 m s^{-1}.

For this wave

(i) Calculate the time period the stationary wave on the string.

[1]

(ii) Calculate the length of the string.

[2]

Question 6 Medium [11 marks]

Part (a) [3 marks]

Fig. 1.1 shows a stationary wave formed on a guitar string fixed at **P** and **Q** when it is plucked at its centre.

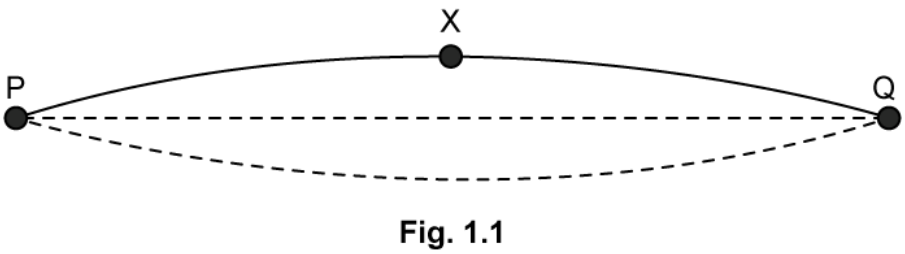


Fig. 1.1

X is a point on the string at maximum displacement.

Explain why a stationary wave is formed on the string

Part (b) [3 marks]

The stationary wave in Fig. 1.1 is the D string of the guitar which has a frequency of 146.83 Hz.

Calculate the time taken for the string at point X to move from maximum displacement to its next maximum displacement.

Part (c) [2 marks]

The progressive waves on the string travel at a speed of 190 m s^{-1}.

Calculate length of the D string.

Part (d) [3 marks]

A guitarist presses on the string at point R to shorten it and create the higher note 'E'. The distance between R and Q in Fig. 1.2 is 0.29 m.



Fig. 1.2

The speed of the progressive wave remains at 190 m s^{-1} and the tension remains constant.

Calculate the frequency of note E.

Question 7 Hard [12 marks]

Part (a) [2 marks]

Sound waves, like waves on a string, can produce stationary waves inside air columns.

A physics technician demonstrates how sound can set up stationary waves using a tall tube of cross-sectional area 4.0×10^{-3} m^2 and a loudspeaker connected to a signal generator, as shown in Fig. 1.1.

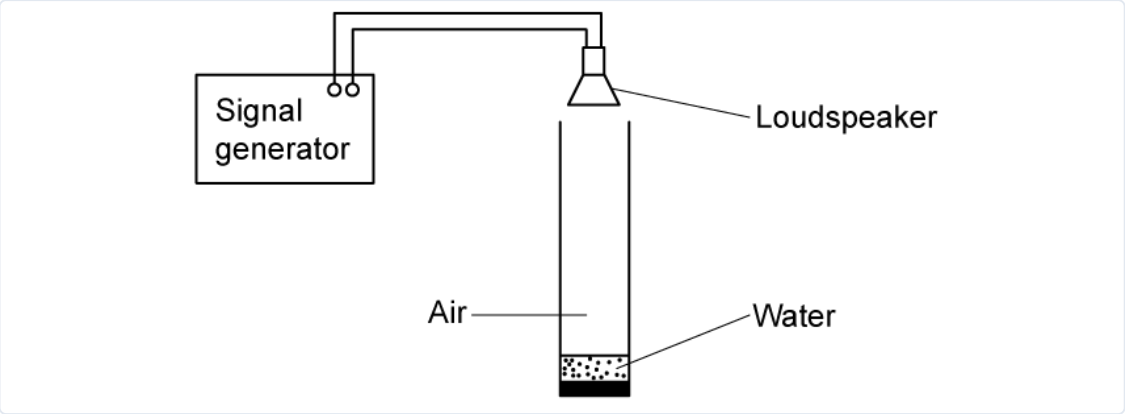


Fig. 1.1

The signal generator is switched on so that sound waves enter the tube. He slowly pours water into the tube and the sound gradually increases in volume, reaching a first maximum at a particular instant. Immediately after the volume begins to decrease.

Water continues to be added until the volume rises again, reaching a second and final maximum after a further 3.2×10^{-3} m^3 of water is poured in.

Determine the wavelength of the sound waves.

Part (b) [3 marks]

To help explain the demo, the technician sketches how particles of air move around in a tube when a stationary sound wave is set up. This sketch is shown in Fig. 1.2.

A displacement anti-node is the maximum movement of particles in the air column. A displacement node is where there is no movement of particles in the air column.

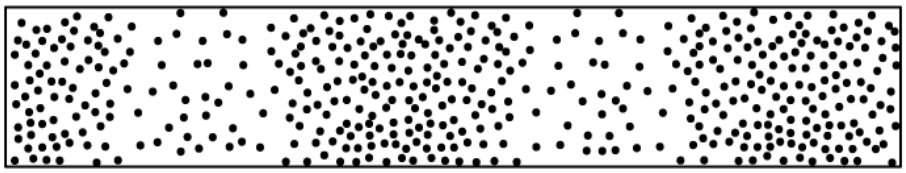


Fig. 1.2

On Fig. 1.2, label all positions where there are displacement nodes.

Part (c) **[3 marks]**

When air passes through a displacement node it is squeezed through the node and it expands away from it, so the change in pressure is at a maximum.

One of the technician's students asks, "Are positions of displacement nodes the same as positions where the air pressure is maximum?"

With reference to the meaning of a displacement node, and using Fig. 1.2, suggest how the technician might respond to the student.

Part (d) **[4 marks]**

In the space provided in Fig. 1.3 below, sketch the shape of the stationary sound wave set up in Fig. 1.2

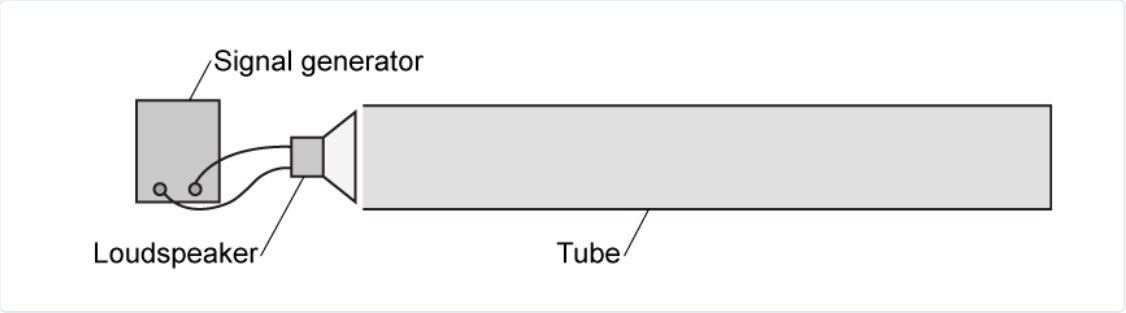


Fig. 1.3

Question 8 Hard [16 marks]

Part (a) [3 marks]

The diagram shows a common piece of laboratory equipment used to demonstrate wave phenomena.



Explain how waves from the loudspeaker form stationary waves in the tube. Include in your answer a condition for the formation of the wave and describe the wave which is formed.

Part (b) [5 marks]

Construct a three-part diagram linking the wave shape, node formation and pressure differences within the tube, for the third harmonic of the wave formed. Start with the template below in Fig. 1.1,



Fig. 1.1

Part (c) **[4 marks]**

The speed of sound in the tube is 340 m s^{-1} and the frequency of the sound emitted by the loudspeaker is 920 Hz for the third harmonic of the wave.

For this equipment calculate:

(i) The length of the tube, giving the answer in cm.

[2]

(ii) The wavelength of the seventh harmonic, giving the answer in S.I. units.

[2]

Part (d) **[4 marks]**

A pipe of the same length has its closed end removed to create a pipe open at both ends.

Show that the seventh harmonic of sound is also produced for sound of the same wavelength as the seventh harmonic when one end of the pipe is closed.